

Balance It

What a coefficient actually does to an equation, what a subscript refuses to do, and why “the atoms match” is only half of the test.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/balance-it/

1 Before you open anything

Answer from what you already know. Being right is not the point — having something on paper to argue with later is.

PREDICTION

A reaction equation contains the term **3 Ca(NO₃)₂**. How many *oxygen atoms* does that term represent? Write one number.

Write one sentence saying how you got it, including which numbers you multiplied.

2 Reading the four places a number appears

Open the demonstration. It loads on reaction 1 of 12, **Hydrogen burns to water**. The equation strip has a – / + box above each formula; those are the coefficients, and they are the only numbers on the page you can change. Under the strip is the **Atom count** table, and in the right-hand column are four boxes under **Where you stand**.

1. Press **All back to 1** so every coefficient reads 1.
2. Fill in the first row of the table below: the **Left** and **Right** entries for Hydrogen and Oxygen come from the Atom count table, and the last three come from **Elements matched, Atoms left** and **Atoms right**.
3. Use the + and – buttons to set each remaining row of coefficients, and record the same six numbers each time.

Coefficients (H ₂ , O ₂ , H ₂ O)	H left	H right	O left	O right	Elements matched	Atoms left	Atoms right
1, 1, 1							
2, 1, 1							
1, 2, 2							
2, 1, 2							

a. One row has **Atoms left** equal to **Atoms right** and is still not balanced. Which row is it, and which elements are wrong in it?

b. From that row, state in one sentence why the two total-atom boxes are not the test for a balanced equation.

c. With the last row still set, press **Check** and copy the bold headline of the message you get.

The formulas themselves have no controls. There is no way to turn H₂O into H₂O₂ on this page, because there is no such move in chemistry either: the formula of a compound is settled before the equation is written. H₂O₂ is a different substance, and it has its own reaction in the picker.

3 What one coefficient multiplies

Choose **Lead(II) nitrate and potassium iodide** from the reaction menu, then press **All back to 1**. Change nothing except the coefficient on **Pb(NO₃)₂**: leave the other three at 1 for this whole section.

1. With the coefficient on Pb(NO₃)₂ at 1, read the **Left** column of the Atom count table for Lead, Nitrogen and Oxygen, and read the **Atoms left** box.
2. Press + once. Record the same four numbers.
3. Continue up to a coefficient of 5, one press at a time.

Coefficient on $\text{Pb}(\text{NO}_3)_2$	Lead, left	Nitrogen, left	Oxygen, left	Atoms left
1				
2				
3				
4				
5				

a. How many Pb, N and O atoms are in *one* formula unit of $\text{Pb}(\text{NO}_3)_2$? Get each from your table, not from the formula.

b. Write the rule in your own words: given a coefficient c , a subscript inside the bracket and a subscript outside it, how do you get the atom count for that element?

c. Going from coefficient 1 to coefficient 3 tripled Lead, Nitrogen and Oxygen, but **Atoms left** did not triple. Explain why not.

d. Go back to your prediction in part 1. Your row for coefficient 3 gives the oxygen count for 3 $\text{Pb}(\text{NO}_3)_2$, which has the same bracket structure as 3 $\text{Ca}(\text{NO}_3)_2$. Was your prediction right? If not, name the multiplication you left out.

4 When “every row is green” is not enough

Select **Calcium nitrate and sodium phosphate** and press **All back to 1**. Balance it yourself. Counting nitrate and phosphate as whole units, rather than counting oxygen atom by atom, will save you a great deal of arithmetic. Do not press *Show the answer* — reactions you reveal do not count towards the **Balanced so far** total.

a. Write the balanced equation with the coefficients you found.

b. Record these from the screen once it balances, then press **Double every coefficient** and record them again.

	Coefficients (4 numbers)	Oxygen left	Oxygen right	Elements matched	Atoms left
Your balanced answer					
After doubling					

c. Press **Check** on the doubled version. Every row of the Atom count table is still green, yet the check rejects it. Copy the bold headline, then state the second condition the check is testing.

d. A shortcut worth distrusting: “if the atoms match, the equation is finished.” Give the arithmetic test that decides whether a set of matching coefficients is the accepted answer.

e. Oxygen appears in nitrate on one side and in phosphate on the other, and the totals still have to agree. Explain what the balanced equation does and does not claim about where those oxygen atoms end up.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working — carbon and metals first, hydrogen next, oxygen last. Then reopen the demonstration, build each one, and if you were wrong write down exactly where.

1. Balance, in lowest whole-number terms: $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

2. Balance, in lowest whole-number terms: $\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$. Balancing carbon and hydrogen first leaves an oxygen count that cannot be reached with a whole coefficient on O_2 . Say what you did about that, and why it is allowed.

3. Someone submits $4 \text{KClO}_3 \rightarrow 4 \text{KCl} + 6 \text{O}_2$. Check each element and say whether the atoms are conserved. Then say whether the answer would be accepted, and write the form a chemist would write. Explain the difference between the two questions you just answered.

4. For $3 \text{Ca}(\text{NO}_3)_2 + 2 \text{Na}_3\text{PO}_4$, find the total number of oxygen atoms on that side. Show the two separate contributions, and state why the answer is not $3 \times 3 = 9$.

6 On your own

Nothing here is graded. The demonstration holds twelve reactions and this worksheet used four of them.

- Take **Aluminum corrodes in air** and try to balance it with an *odd* coefficient on Al_2O_3 . Work out from the atom table why no odd value can ever work, then find the answer.
- Watch the **Particle view** while you press +. Nothing about any drawn unit changes shape; another identical copy appears. Write one sentence connecting that to why a coefficient can never break a substance apart.
- Work the two single-replacement reactions, **Zinc in hydrochloric acid** and **Aluminum in copper(II) chloride**, and describe what the second one demands that the first one does not.
- Drive **Balanced so far** to 12 / 12 without using *Show the answer* once.